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Module 2

14/01/26 Biometric standards ensure:-

1) Interoperability
2) Data exchange
3) Performance evaluation

Biometric standards ensure interoperability, quality and security across diff biometric s/m & tech

Key Organisations

- ISO/IEC JTC 1/SC 37
- NIST
- ANSI
- IEEE

ISO/IEC Biometric stds

ISO/IEC 19794 - Biometric data interchange formats for fingerprint, face, iris and other modalities.
ISO/IEC 19795 - Biometric performance testing & reporting stds.
ISO/IEC 24745 - Biometric info protection stds
ISO/IEC 30107 - presentation attack detection stds.

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Fingerprint

ISO/IEC 19794-2: fingerprint minutiae data format
ISO/IEC 19794-4: fi image data format
ANSI/NIST-ITL 1-2011: Data format for exchange of fingerprint, facial & other bio info
FBI EBTS: Electronic biometric transmission specification
WSQ: wavelet scalar Quantization compression std

Face

ISO/IEC 19794-5: face image data interchange format
39794-5: updated face image data std
ICAO 9303: Machine readable travel doc
ISO/IEC 29794-5: face image quality stds
IEEE 2410: Biometric privacy framework including facial recog

Iris Recog

ISO/IEC 19794-6: Iris image data
ISO/IEC 29794-6: Iris image quality
ISO/IEC TR

Voice & other

Voice recog

ISO/IEC 19794-13: Voice data interchange
ISO/IEC 30115: Speaker recognition vocabulary.

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Other

ISO/IEC 19794-8: Finger pattern spectral data

7: Signature/sign behavioural data

8:

Application-Specific

Border control

ICAO 9303 - Machine readable travel doc

ISO/IEC 7816 - Smart card integration

Mobile Biometric

IEEE 2410-2021 - Bio privacy

FIDO Alliance - Authentication stds

Healthcare

ANSI INCITS 358

Quality and performance stds

Quality Assurance

ISO/IEC 29794 series: Quality framework for diff modalities

NIST NFIQ: fingerprint image quality std

Performance testing

ISO/IEC 19795-1: Principle & framework

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ISO/IEC 19795-2: testing methods

ISO/IEC 19795-7: testing of on-card biometrics

Security & Privacy Std

ISO/IEC 24745: Biometric info protection framework

ISO/IEC 30107 series: presentation attack detection

ISO/IEC 27001: info security management

Emerging biometric std

Behavioural biometric: Gait, keystroke dynamics, mouse movement pattern

multimodal s/m: ISO/IEC 24779 series from cross-jurisdictional s

AI & ML: stds for neural n/w-based biometric

continuous authentication: std for ongoing identity verification

Biometric encryption: template protection & cancellable biometrics

Apply ISO/IEC 24745 and privacy framework to biometric s/m and the case study provided are:

i) workplace monitoring employee presence

ii) law enforcement database

iii) health care patient identification

a) Identify privacy risk

b) map to ISO/IEC 24745

c) Assess local privacy law

d) Propose privacy enhancing technologies

e) design data protection measures
f) create consent & transparency mechanism.

Ans: ISO/IEC 24745 provides guidelines to protect biometric info by ensuring:
- Data minimization
- Purpose limitation
- Irreversibility
- Unlinkability
- confidentiality
- Revocability and accountability

i) workplace monitoring employee presence
a) Identify Privacy Risks
- continuous surveillance of employees
- Function creep (attendance data used for performance monitoring)
- unauthorized access to biometric data
- Insider misuse by administrators
- Data breaches leading to identity theft.
- Lack of informed consent.

b) Map Privacy Risks to ISO/IEC 24745 Requirements

Privacy Risk	ISO/IEC 24745 principle applied
Over-collection of data	Data minimization
use beyond attendance	Purpose limitation
Identity theft risk	Irreversibility of biometric templates
Linking attendance with HR records	Unlinkability
insider misuse	confidentiality & access control
No control for employee	Revocability

c) Assess Local Privacy law
- Digital Personal data Protection Act (DPDP), 2023
- Biometric data classified as sensitive personal data
- Requires:
  - Explicit employee consent.
  - Purpose limitation
  - Secure processing
  - Right to withdraw consent.

d) Propose Privacy Enhancing ~~technologies~~ techniques
- cancelable biometrics
- Encrypted biometric templates
- On-device biometric matching
- Tokenization of biometric identifiers.

e) Design data Protection measures
- Encrypted storage of biometric templates.
- Role-based access control
- Secure audit logs
- Automatic deletion after employment ends
- Periodic security audits.

f) create consent & transparency Mechanism
- written consent during onboarding
- clear privacy notice explaining usage
- Alternative attendance method
- Employee access to attendance records
- complaint and grievance mechanism

ii) Law enforcement database
a) Identify Privacy risks
- Mass surveillance
- False identification and wrongful arrest
- Misuse of data beyond law enforcement
- Data breaches
- Lack of individual consent

b) Map to ISO/IEC 24745 requirements

Privacy Risk	ISO/IEC 24745 principle applied
excessive data collection	Proportionality & minimization
wrongful identification	Data accuracy & integrity
Misuse of biometric data	Purpose limitation
Database hacking	Confidentiality & secure storage
Tracking individuals	Unlinkability

c) Assess local Privacy law
- DPDP act allows exemptions for:
  - Law enforcement
  - national security
- still requires:
  - safeguard against misuse
  - Accountability
  - Oversight mechanisms.

d) Propose Privacy enhancing ~~technologies~~ techniques
- Biometric cryptosims
- one-way template protection
- secure multi-party computation
- Threshold-based matching to reduce false positives

e) Design data Protection Measures
- Isolated and encrypted database
- Multi-factor authentication for officers
- Access logging and monitoring
- Periodic audits by independent authority
- Data retention limits

f) create consent & transparency mechanism
- Legal authorization for data collection
- Public transparency reports
- Right to correction and deletion for innocent persons
- Judicial oversight of database access

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iii) Healthcare Patient Identification

a) Identify Privacy Risks

- Exposure of biometric and health data
- unauthorized access by hospital staff
- Data sharing with insurers w/o consent
- Re-identification risks
- Medical identity theft

b) map to ISO/IEC 24745 requirements

Privacy Risk	ISO/IEC 24745 principle applied
Sensitive data exposure	Confidentiality
Excessive data linkage	unlinkability
Permanent identifiers	Revocability
Unnecessary collection	Data minimization
Data misuse	Purpose limitation

c) Assess local Privacy Laws

- DPDP Act, 2023
- Medical and biometric data = highly sensitive
- Requires:
  - Explicit informed consent
  - Strong safeguards
  - Breach notification

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d) Propose Privacy enhancing technigues

- Biometric tokenization
- Secure enclaves
- Pseudonymization
- Federated identity sms

e) Design data Protection measures

- separate biometric and medical databases
- End-to-end encryption
- Limited staff access
- Regular backups and monitoring
- Anonymization for research use

f) Create consent & transparency mechanisms

- informed consent at patient registration
- clear explanation of biometric usage
- Right to access and correct data
- Opt-out where feasible
- Display privacy policy in healthcare facilities

20/01/26 Biometric Sim Architecture

1. Sensor Module
2. Single processing
3. Feature extraction
4. Template storage
5. Matching module

Biometric Modes

1. Enrollment mode
2. Verification mode (1:1)

ROC curve analysis

→ Receive operating characteristic (ROC) curve shows the trade off b/w FAR and FRR at different threshold values
- High threshold: structure matching → low FAR, high FRR (more secure, less convenience)
- Low threshold: lenient matching → high FAR, low FRR (less secure, more convenient)
- Optimal point: balanced based on appl^n needs

Appl^n eg
high security (nuclear facility): needs to be highly secure
convenience (cryo access)

Design process

1. Step 1

→ Application analysis
- purpose (access control, time attendance)
- Scale (no. of usr, 100s, 1000, millions)
- Environment (indoor, outdoor, mobile, fixed)

→ user population characteristics
- Demographics (age range, phy abilities)

- user cooperation levels (cooperative, non-cooperative)
- Technical literacy.

→ Security Requirments
- Acceptable FAR (how many false accept tolerance)
- Acceptable FRR - (how many false rejects, tolerance)
- attack ~~distance~~ ~~dista~~ resistance needed.

2. Step 2 (Check characteristics)

criterion	Description

Biometric Modality Comparison

Modality	Best use case	Accuracy	cost	Acceptability
fingerprint	general purpose, forencics	high	low	high
iris	high security, border control	very high	high	medium
face	surveillance, mobile devices	medium high	low medium	very high
voice	remote authentication, call center	medium	low	high
hand geometry	physical access control	medium	medium	high
signature	document verification	low medium	low	very high

Multimodal biometric system
combining multiple biometric traits or capture methods to improve performance.

Levels
sensor level - multiple sensor for same trait
feature level - combine feature from different sources
score level - combine matching scores
decision level - combine final accept/reject decision.

Advantages
- high accuracy, reduced FAR & FRR
- Increased population coverage: Secure for s/m with poor quality in one trait
- Spoof Resistance - Harder to fake multiple replicas.
- Flexibility - use different traits for different security level.

→ Algorithm and database design
Matching algorithm
→ correlation based
- Direct image comparison using correlative coefficient.
- Simple but computationally exp

→ Feature based
- Compares extracted features
- More efficient and robust

→ Database consideration
- Scalability performance

- Security : Template protection
- Redundancy

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→ case study
- Smartphone unlock features
Design a biometric solution for following.
1) which biometric modality to use
Ans: Face recognition and fingerprint.
2) Enrollment process flowchart
Face recognition.
a) Data capture - user's face is captured using sensor (phone camera)
↓
b) quality check - System verifies if the captured data is sufficient for recognition
↓
c) Registration - Data is processed and stored in data base

3) Verification/identification process
Ans: user scan their face infront of phone camera. s/m verifies it with stored preformed template. if it is greater than threshold. access given, else denied. → feature extraction, template storage, matching
4) Security measures against spoofing
Liveness detection - check eg blinking to prevent mask/pic spoof
5) Handling failure case (FAR - FRR trade off)
- Allow retry, back up option

* Security threats and counter measures
- Spoofing (fake biometric) liveliness detection multimodal
- Replay attack (resubmit prev captured) · timestamp verification, session token
- Template attack (steal stored templates) : encryption template protection schemes.
- Channel attack (intercept data b/w modules) : secure communication protocols
- Jagan Arase

Liveliness detection techniques
- fingerprint : pulse detection, temp sensing, blood flow.
- face : 3D depth mapping, thermal imaging
- iris : pupil dilation response to light
- Texture analysis detect difference b/w real skin & fake materials
- machine learning : Train classifier
- Distortion analysis

22/01/26 Biometric Matching process
1. capture biometric sample from sensor
2. Pre process the sample
3. Extract feature
4. Retrieve stored template
5. Apply matching algorithm

Classification of Matching Methods

	Correlation based	Feature based
Approach →	direct comparison of biometric image/signals	extract and compare distinctive features
Method →	pixel to pixel / signal correlation eg: cross-correlation, normalized-correlation, phase correlation Best for raw image matching, retina patterns	feature vector comparison eg: minutiae matching (fingerprint) Landmark matching (face) Texture pattern matching (iris) Best for most biometric modalities.

Distance matrices for matching
- distance matrices measure dissimilarity b/w templates
- lower distance = higher similarity.

1. Euclidean distance
$d(P,Q) = \sqrt{\Sigma(P_i - q_i)^2}$
- most common
- measure straight line distance in feature space

2. Manhattan distance (L_1 Norm)
$d(P,Q) = \Sigma|P_i - q_i|$
- sum of absolute diff
- less sensitive to outliers than euclidean

3. Manhattan distance (L_1 Norm)

$d(P,Q) = \Sigma|P_i - q_i|$

Sum of absolute difference

Less sensitive to outliers than Euclidean.

3. Hamming distance

for $d(P,Q)$ = no. of positions where bits differ.

- used for binary feature vectors (eg: iris code)
- very fast computation.

4. Mahalanobis distance

$d(P,Q) = \sqrt{((P-Q)^T \Sigma^{-1} (P-Q))}$

Accounts for correlation b/w features & different scales.

$\Sigma \rightarrow$ Covariance matrix

5. Cosine similarity

Similarity $= (P.Q) / \|P\| \times \|Q\|)$

measures angle b/w vectors

Value ranges from -1 to 1 (1 = identical direction)

6. Chi-square Distance

$d(P,Q) = \Sigma((P_i - q_i)^2 / (P_i + q_i))$

- used for histogram-based features (eg: texture descriptions)

Fingerprint Matching Method

→ Advantage:

- Small template size
- widely standardized
- works with partial parts
- rotation / translation invariant

→ Disadvantage:

- sensitive to image quality
- minutiae detection errors
- Alignment complexity
- performance degrades with dry/wet finger.

1. Minutiae - Based Matching

- Ridge endings and bifurcations
- set of (x, y, θ) coordinates for each minutiae point.

Matching Algorithm

1. Align templates
2. Find corresponding minutiae pairs
3. count matched minutiae pts
4. compute match score = (matched pairs) / (total minutiae)

2. Correlation Based : Direct comparison of fingerprint image using correlation based

$[\Sigma\Sigma I_1(x,y) \times I_2(x,y)] / \sqrt{(\Sigma\Sigma I_1^2 \times \Sigma\Sigma I_2^2)}$

3. Ridge feature Based
- Ridge orientation : direction of ridges at each pt.
- Ridge frequency : no of ridges per unit distance.
- Ridge shape : curvature and flow patterns.

4. Hybrid method - combine minutiae + texture features for improved accuracy.

comparison table

Method	Accuracy	Speed	Storage
Minutiae	high	Fast	Small
correlation	very high	slow	Large
~~Ridge feature~~ Hybrid	v. high	medium	medium

Face Recognition Matching Methods

1. Geometric / feature based Extract facial landmarks (eye, nose, mouth) and compute geometric relationships.

feature:
→ distance b/w eyes
→ nose width & length
→ face width & height
→ Jaw line shape
* matching - Euclidean distance on feature vector.

2. Appearance based (holistic) : Treat entire face as a pattern.
2 sub-methods :
1. eigenfaces (PCA)
- Project faces onto principal components
- match: euclidean distance in eigenspace.
2. fisherfaces (LDA)
- maximize class separability
- match : Mahalanobis distance

Modern Face recognition - deep learning

3. deep learning - based matching
1. Face detection - Locate face in image
2. Face alignment - normalize pose & scale.
3. CNN feature Extraction - 128-512 dimension embedded
4. Matching - cosine similarity / euclidean distance

popular Architecture
* Face Net - triplet loss learning for embeddings
* Deep Face - 9 layer deep n/w by facebook
* VGG Face - based on VGG-16 architecture
* Arc Face - additive angular margin loss

Advantage : Achieves >99% accuracy on standard benchmarks

Iris Recognition matching

Daugman's Algm (most popular)

1. Iris segmentation - detect inner & outer boundaries (separate iris from eye in image captured)
2. Normalization - convert to ~~rean~~ rectangular format (polar coordinates)
3. Feature Encoding - Apply 2D gabor wavelets
4. Create Iris code - 2048 bit binary template.

Matching Process

- formula: Hamming dist = (XOR of 2 iris codes) / (total bits)
- Decision: If HD < 0.32, declare match (typically)

→ Binary iris code allow extremely fast matching (millions of comparison per second). XOR operation is hardware-optimized, hence we used HD.

23/01/26 Voice Recognition

Feature Extraction

MFCC: Mel-Frequency Cepstral coeff

LPCC: Linear Predictive coding

Pitch and Formants: Fundamental Frequency chara

Matching

1. Template

DTW (Dynamic time wraping)

- Align sequences of different length

- finds optimal path through distance matrix
- Good for text path dependent system.

2. Statistical Models

GMM (Gaussian Mixture model) - Bell shaped curve

- Model voice as a mixture of Gaussians
- compare likelihood of test sample
- text independent recognition

3. Deep learning (I-vector & x-vector): Extract fixed dimensional embeddings, then use cosine similarity for matching

Signature Verification Matching

Static (offline)	Dynamic (online)
- image of signature on paper	- captured during process
- Features:	- Features
- shape of descriptors	- pen position
- Geometric features	- Pressure & angle
- Pressure pattern	- Velocity & acceleration
- Texture Analysis	- Timing information
- Matching: correlation / feature distance	- Matching: DTW & temporal sequence

Template Alignment Techniques

- Critical challenge: Biometric sample vary due to position, rotation and scale changes.

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1. Brute force search
- Try multiple alligments & select best match
- computationally expensive
- used when other methods fail

2. Reference point alligment
- Detect stable reference pt (core pt in fingerprint)
- Align templates based on reference pt.
- Fast but depends on accurate detection.

3. Iterative closest Point (ICP)
- Initialize transformation
- find nearest neighbours b/w templates
- Estimate optimal transformation
- Apply transformation
- Repeat untill convergence

4. RANSAC (Random sample consensus)
- Robust to outliers
- Randomly sample correspondences & find best transformation.

Score Normalization Techniques

Different matchers produce scores on different scales.
Normalization is essential for:
- Multimodal fusion
- comparison across SIMs
- Quality based weighting

Matching Performance optimization

1. Indexing & Binning: Reduce search space for 1:N identification.
fingerprint: classify by pattern type (loop, whorl, arch)
face: gender, age, ethnicity filters
iris: texture-based clustering

2. Cascaded matching
1: Quick coarse matching to filter candidates
2: Detailed fine matching on filtered subset.
eg: use global features first, then local minutiae details.

3. Hardware Acceleration
- GPU parallel processing for deep learning
- FPGA for real-time iris matching
- SIMD instructions for correlation

Quality based matching

Incorporate quality scores to improve matching decisions.

Image quality	Feature Quality
Resolution	- No. of features
contrast	- Feature reliability
Focus/sharpness	- Occlusion : hindrance in capturing the features
Lighting	- Pose variations
Noise level	- Expression (face)

Quality-weighted Matching

Formula : final_score = match-score × Quality-factor

Strategy:

- Low quality sample → request re-capture
- Medium quality → Adjust threshold
- High quality → use standard threshold

Method	Accuracy	Speed	Mly
correlation	High	slow	High
Minutiae (fingerprint)	High	Fast	Low
Hamming (iris)	V High	V Fast	V. Low
Eigen-face	Medium	Fast	Med
Deep learning	V high	Fast	High
DTW (voice)	Med high	Med	Med

challenges in Biometric matching

Intra-class Variation (same person, different samples)	Inter class variations
- sensor noise	- twins or relatives
- environmental changes	- limited feature space
- user interaction variability	- population strategies
- Aging effect	- similar appearance
- injuries or conditions	- partial data overlap
- pose & expression	

Other

Non-linear distortion

Occlusion

Scalability

Template aging

Practical Example : Fingerprint Matching

Step 1: Enhancement

Apply Gabor filter for ridge enhancement

Binarization and thinning

Step 2: Minutiae Extraction

Detect ridge endings: count neighboring pixels

Detect bifurcations: count ridge branches

Extract (x,y,θ) for each minutiae

Step 3 Alignment
- Detect a reference point (core/delta)
  Rotate and translate query to align with template.
  Try multiple alignment hypotheses if needed

Step 4 Matching
For each template minutia, find nearest query minutiae
count matches within tolerance ($\Delta x, \Delta y, \theta$)
Typical tolerance, 10 pixel, 15 degree

Step 5 Scoring
Match score = Matched minutiae / min(Query_minutiae, template_minutiae)

Step 6 Decision
if match score ≥ 0.65 - MATCH
else: No match

Practical Example: Iris Matching
Input: Query iris image, stored iris code

Step 1 Preprocessing
- Detect pupil and iris boundaries
- Normalize iris region to fixed dimension (512×64)
- Compensate for eyelid/eyelash occlusions.

Step 2 Feature Encoding
- Apply 2D Gabor wavelets at multiple scales
- Phase quantization: 2 bits per pixel
- Generate 2048 bit iris code
- Create noise mask for occluded regions

Step 3 Matching
- Shift query code ± 15 pixels to compensate for rotation
- for each shift position:
  XOR_result = Query Code XOR Template code
  Valid_bits = bits NOT masked by noise
  HD = Count(XOR_result) / valid bits

Step 4 Decision
HD = maximum HD across all shifts
if HD < 0.32 match (genuine)
  HD > 0.45 No match (imposter)
  $0.32 < HD < 0.45$ uncertain - recapture

Security in biometric matching

Cancellable Biometric	Fuzzy vault
• Apply 1-way transformation	• hide template in
• can revoke & reuse	• genuine match unlock vault.
• matching in transformed domain	

Best practices for biometric matching

1. Always normalize scores b/w fusion or comparison.
2. Implement quality checks - reject poor quality samples early.
3. use multiple enrollment samples for robust templates.
4. set thresholds based on appl^n req, not defaults
5. log match scores and decision for performance monitoring.
6. Test with diverse population representing actual users.
7. Implement liveness detection to prevent spoofing
8. Provide clear feedback to users for failed matches.

Why accuracy matters

- no biometric s/m is 100% accurate
- Errors occur due to sensor noise, environmental factors and biological variation
- we need standarized metrics to measure & compare s/m performance
- different appl^n requires different accuracy level.

types of errors

- Type I Error: False match
  - s/m accepts an impostor
- Type II Error: False Non-match
  - s/m rejects a genuine user

Trade-off

- reducing one error typically increase the other.
- s/m designers must balance security vs convenience
- threshold adjustment control this trade off

Basic terminology

Genuine Attempt:	Impostor Attempt:
• comparison b/w sample from same person	• comparison b/w sample from different person.
• eg: user's fingerprint compared against their own stored template	• eg: Attacker's fingerprint compared ~~against~~ against victim's template
• Exp result: should match	• Exp result: should n't match

Match score

Numerical value indicating similarity b/w 2 biometric samples.

- Low score: samples are different (likely diff person)
- High score: samples are similar (likely same person)
- Threshold (τ): decision boundary for match / non match

Confusion matrix

	Predicted match	Predicted Non-match
Actual: Genuine	True match (TM) Correct acceptance	False Non match (FNM) Incorrect rejection
Actual: Impostor	False match (FM) Incorrect acceptance	True Non-match (TNM) Correct rejection

False Match Rate (FMR) / FAR

- probability that the s/m incorrectly accepts an impostor

$FMR = (\text{no. of } FM) / (\text{no. of impostor attempts})$

$FMR = FM / (FM + TNM)$

- Also known as : FAR, type I error
- security metric : lower FMR - high security
- Risk : unauthorized access to s/m / resources
- critical for : high security appl^n (banking, military, border cntrl)

False Non Match Rate (FNMR)

- s/m incorrectly rejects a genuine user
- FNMR = (no. of FNM) / (no. of genuine attempts)
  = FNM / (FNM + TM)
- Also known as : FRR, type II error
- usability metric : ↓ FNMR = Better user experience
- impact : user frustration, denied access to legitimate users.
- critical : convenience appl^n (phone unlock, attendance s/m).

Threshold setting	FMR	FNMR	use cases
Low threshold (easy to match)	high↑	Low↓	gym access, low secure
Medium (Balanced)	Med	Med	office access ~~map~~ mobile phone
high (hard to match)	Low↓	High↑	Bank vault, military facilities.

True Match Rate (TMR)

- s/m correctly accepts a genuine user
- other names - TAR, GAR, sensitivity (in general classification), hit rate.

TMR = TM / (TM + FNM)
= 1 - FNMR

TMR = (no. of true matches) / (no. of genuine attempts)

→ TMR + FNMR = 1

True Non match Rate (TNMR)

- s/m correctly rejects an imposter
- other names → TRR, specificity
- TNMR = no. of true NM / no. of imposter

TNMR = TNM / (TNM + FM) = 1 - FMR

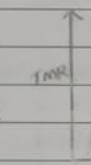
TNMR + FMR = 1

Equal Error Rate (EER)

pt where FMR = FNMR

ROC curve

- Better s/m : curve closer to top left corner
- each pt : rep performance at diff threshold
- Area under curve (AUC) : overall performance measure (close to 1.0 is better)
- operating pt : choose based on appl^n req



Y-axis : TMR
- want high.

x-axis : FMR
- want low.

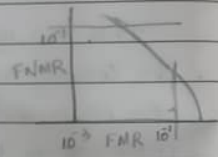
DET curve (Detection Error Trade off)

* DET curve: It shows the trade off b/w security errors (FMR) and usability errors (FNMR)
  - Both axes are in logarithmic scale to classify visualise small error rates

Better Bio S/m:

→ curve lies closer to origin (0,0)

→ Indicates low FMR & low FNMR simultaneously



Feature	ROC	DET curve
Axes	TMR vs FMR	FNMR vs FMR
Scale	Linear	logarithmic
Better	top left corner	origin (bottom-left)
use	genuine comparison	low error rate comparison

FTE	FTA
user who can't be enrolled in s/m	s/m who cannot capture Biometric sample
FTE = (failed_enrollment) / (total_enrollment attempts)	FTA = (failed_acquisition) / (total_acquisition attempts)
cause:	cause:
• poor quality bio	user interaction error
• phy disability	

Accuracy & Error rate

Accuracy

Overall correctness of biometric s/m

Accuracy = (TM + TNM) / (TM + TNM + FM + FNM)

= correct decisions / total attempts

Error rate

Error rate = 1 − Accuracy

= (FM + FNM) / total

Why not just use accuracy?

- It can be misleading when data is unbalanced

eg. 99% attempts are genuine, s/m that accepts everyone has 99% accuracy but terrible security.

Precision: +ve predictive value

= TM / (TM + FM)

- Focus: Reliability of match decision

→ high precision needed
↳ when false matches are costly (security appln)

Recalls: same as TMR / TAR

= TM / (TM + FNM)

- Focus: coverage of genuine users

→ high recall needed
↳ when missing genuine users is problematic (user convenience)

F-score (F-Measure)

F_1-score: harmonic mean of precision & recall

= 2 × (Precision × recall) / (Precision + recall)

= 2 × TM / (2 × TM + FM + FNM)

generalized F-score

$F_\beta = (1+\beta^2) \times (\text{precision} \times \text{recall}) / (\beta^2 \times \text{precision} + \text{recall})$

$\beta = 1$: F_1 score (equal weight to P & R)

$\beta < 1$: more weight to P

$\beta > 1$: more weight to R

d-prime (d')- decidability

statistical measure of separation b/w genuine and imposter distance.

$d' = |\mu\text{-genuine} - \mu\text{-imposter}| / \sqrt{(\sigma^2\text{genuine} + \sigma^2\text{imposter})/2}$

μ-genuine - mean of genuine match score

μ-imposter - " imposter

σ^2-genuine - std deviation of genuine scores

σ^2-imposter - " imposter

d' value	Interpretation	S/m quality
$d' < 1$	high overlap b/w distribution	Poor
$1 \le d' < 2$	moderate separation	Fair
$2 \le d' < 3$	good "	Good
$d' \ge 3$	excellent "	Excellent

decidability index (D)

Alternative measure of distribution separation.

Aspect	d-prime	D
denotes	Avg variable	sum of var
sensitivity	less sensitive to unequal var	more sensitive to outlier
usage	more common	

AUC

probability that a randomly chosen genuine score is higher than a randomly chosen ~~ten~~ imposter score.

$0 \le AUC \le 1$

AUC = 1.0 : perfect classifier (100% discrimination)

AUC = 0.9 - 0.99 : excellent performance

AUC = 0.8 - 0.89 : Good

= 0.7 - 0.79 : Fair

= 0.5 Random guessing (no discrimination)

< 0.5 worse than random (s/m is inverted)

Advantages of AUC

- Threshold independent metric
- Single no. for overall performance
- Accounts for all possible operating pts
- useful for comparing different s/ms.